

Notebook 3c Interpretations

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This document serves as a very rough and unofficial write-up of some findings in uncertainty quantification through Notebook 3c on the SBI NPE normalizing flows models created by Edward. The following are the details of the model created in Notebook 2c, which is being interpreted in Notebook 3c and this document reports with respect to this model.

Test RMSE	Test R^2	Test MAE	Val RMSE	Val R^2	Val MAE
0.103	0.9426	0.0637	0.0944	0.9526	0.0586

Table 1: Model Performance Metrics

Input	Depth	Width	Activation	LR	Flow	Optimizer	Batch	Epochs
Non-PCA 110 Coeffs.	2	16, 16, 8	ReLU()	5.00×10^{-4}	maf	Adam	64	86

Table 2: Model Architecture and Hyperparameters

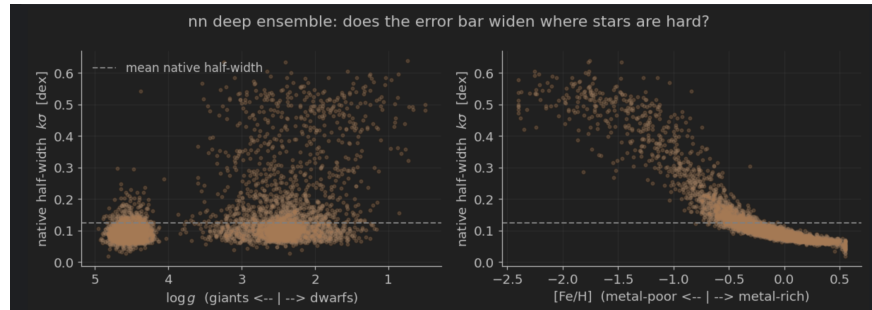
Additionally, the normalizing flow NN (separate from the embedding net) was calibrated to perform 8 transformations and has a 64 hidden features.

1 Error Bars and Uncertainty

The sigma value was calculated by using $\sigma = (q_{84} - q_{16})/2$ for the quantiles q_n for $n = 16, 84$ (the 16th and 84th quantiles). This calculation was done instead of purely using standard deviation as σ in order to account for any tail-heaviness and skews in the posterior distributions that arise from the normalizing flow transformations, which pure standard deviation might not be able to account for accurately since it assumes a normal distribution. In most cases, the posterior does follow a Gaussian-like distribution, but is quite tail-heavy on more extreme metal-rich or metal-poor stars. For further clarity, the 16th and 84th were chosen specifically because they will make up 68.3% of the data ($84 - 16 = 68$) which is how much 1σ in a Gaussian distribution typically covers.

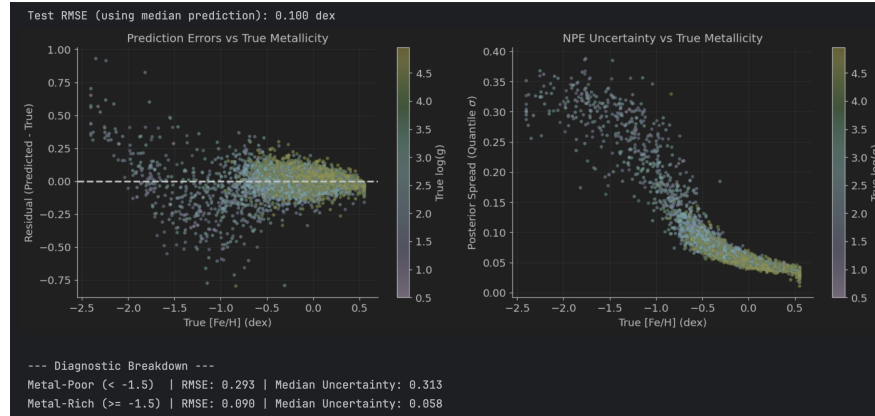
The reported quantile σ values are:

Mean σ	Median σ	Min σ	Max σ
0.076 dex	0.058 dex	0.011	0.388

Table 3: Quantile σ Values

In the right plot above, we see that there where the data is quite condensed (the metal-rich stars), the model is quite certain in its predictions by having a relatively low sigma value. For those stars that are metal-poor, we can see that the model becomes increasingly uncertain, which is quite natural as the amount of metal-poor stars in the dataset are vastly outnumbered (visually) by the metal-rich stars.

Additionally, the left plot shows that most of these uncertainties stem from dwarf stars, which is intuitive as dwarf stars are more common than giant stars given their lifespans. As for why some dwarf stars are vastly more uncertain than others (top of the plot) is an unknown variable to me at this time.

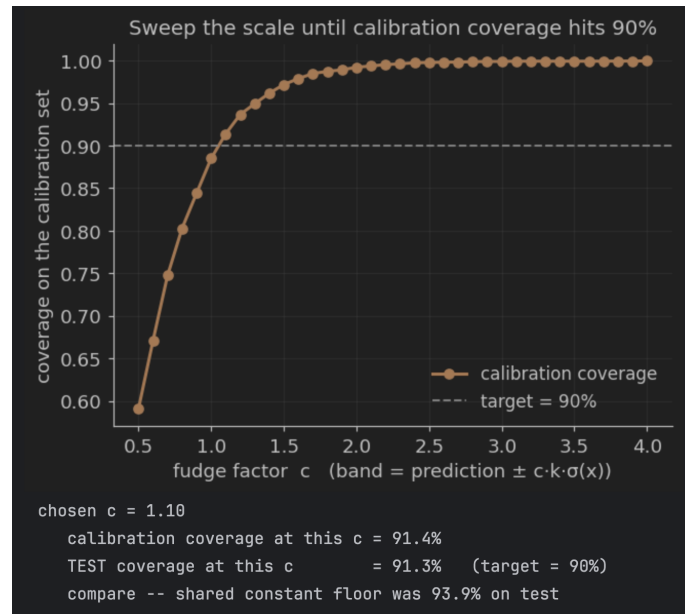


Here is another set of plots, with the right plot being the same as the right plot in figure 1 just with a colouration of the data points according to $\log(g)$. The left plot shows that the model is able to create quite accurate predictions for the majority of stars, however it is underestimating the **FeH** dex value for a certain portion of metal-poor stars, and extremely-poor metal stars are overestimated. This is possibly because the model is generally attempting to trend/bias towards a specific dex interval for metal-poor stars, which produces a slight bit of binomiality in the posterior distributions (if we were to plot a distribution of the means of every distribution).

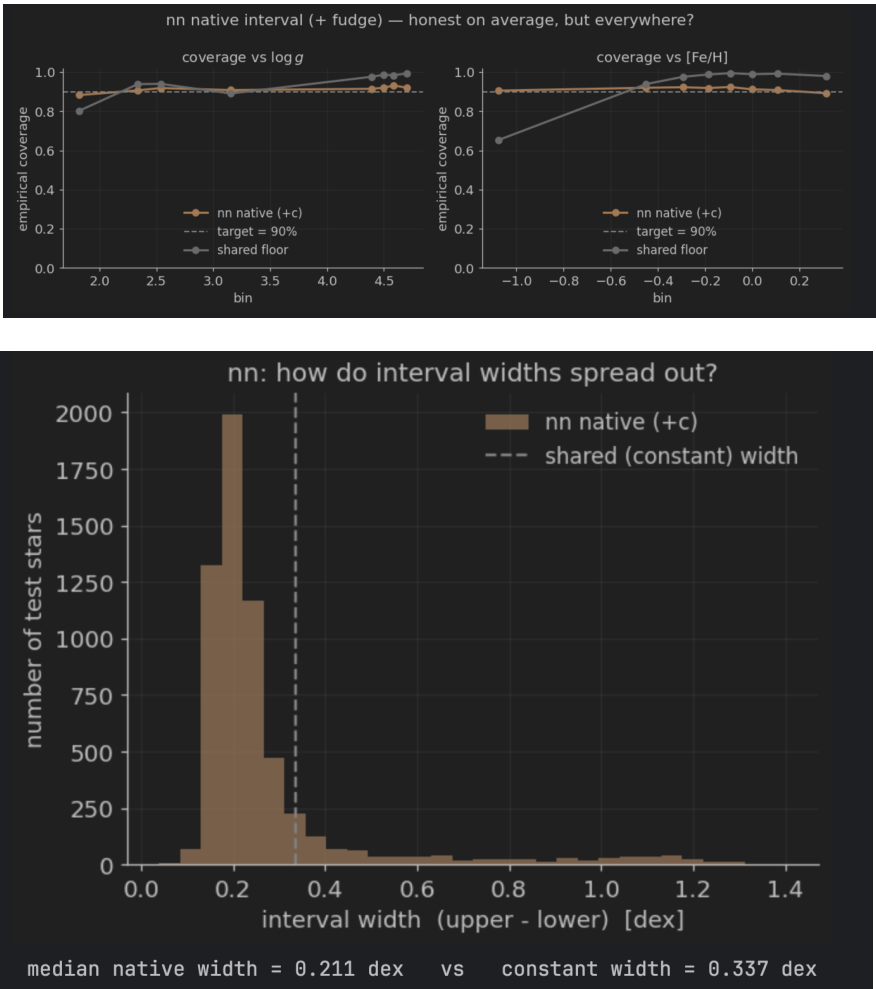
Essentially, I hypothesize that if the means of every posterior distribution were to be plotted, we would see two sections; one sitting in the metal-poor dex range, and one in them metal-rich dex range. However, this is not relevant at the time of writing and will be explored if it is of interest!

2 Calibration Coverages and Interval Widths

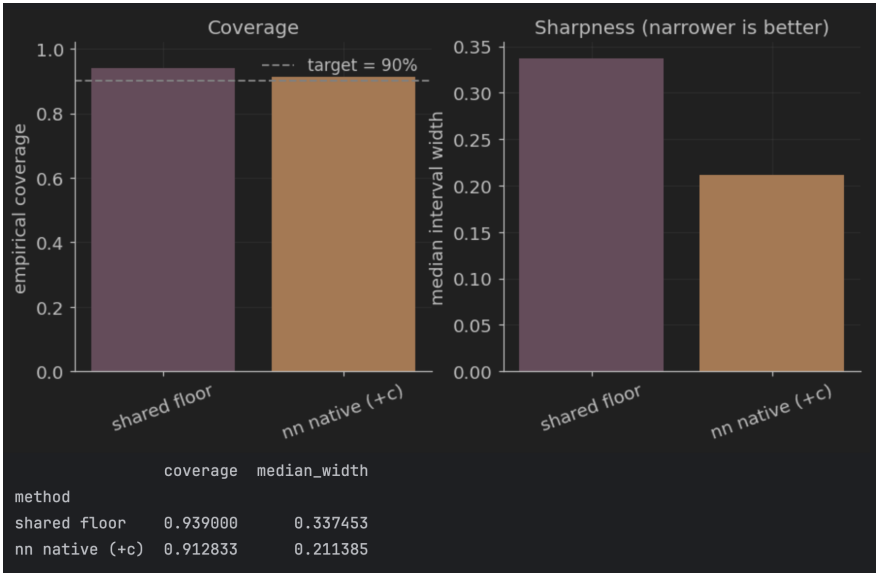
The following are standard plots that came with the notebooks that show where the fudge factor c should be calibrated to, alongside what the model coverage is like with and without the fudge factor being applied.



Plot 2: Chose $c = 1.1$ as the fudge factor to reach the target 90% coverage across calibration and test sets



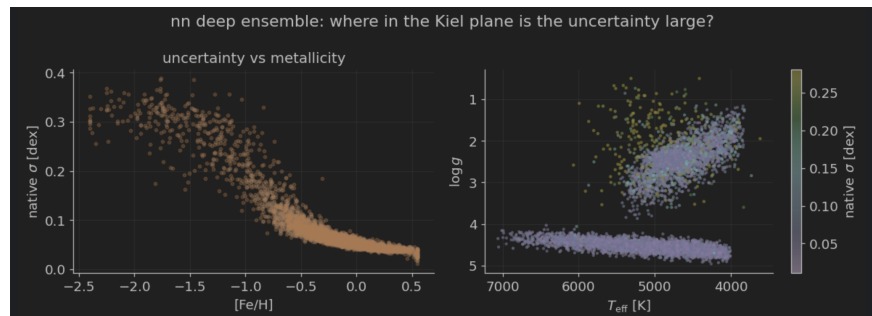
Plot 4: Shows how dense the intervals are for certain dex ranges. Most stars are within 0.4 dex of each other.



Through discussion with Johan and Umang, all of our models follow a similar trend. Good coverage, with the shared floor being a little higher. However, the median interval width is much tighter across our models (orange bar), which simply means our models are condensing where the predicted dex values are. This is aligned with the uncertainty value plots made earlier, as you may see that the majority of the stars are condensed into a low-uncertainty setting.

3 Kiel Plane Uncertainty

Finally, we will look at the plot where the Kiel plane (right plot) shows higher uncertainty.



Interestingly, the heavier stars have quite a low uncertainty overall compared to lighter, cooler stars (dwarves). This is possibly because most of the data is dwarf stars so it's only natural to have a slightly higher uncertainty there, but that's also counter-intuitive because the more data we have, the more certain our model should be. The left plot is simply a repeat of the other uncertainty plots made earlier.

4 Aleatoric or Epistemic Uncertainty

Finally, what is my interpretation of the type of uncertainty present? I'm quite certain that there is a form of epistemic uncertainty present in my model configuration, as there are certain sections of data that simply have higher uncertainty than others which is due to the lower amount of data available for them, and the model struggling to perfectly predict on them. An argument could be made for it to be aleatoric as well, as throughout my many runs of attempting to create models, the best ones were hovering around the 0.07-0.08 validation RMSE range with 95-96% R^2 explained variance. So perhaps the limiting factor simply is that a particular subset of the data is unable to be appropriately trained and predicted upon *because* of the lack of data for that subset (e.g. the metal-poor stars).